

Fourteen Caption Users

AP Statistics · Unit 3 · Topic 3.2 · TEACHER KEY

CED TETHER

- **Skill 3.B** — Determine parameters for probability distributions.
- **Skill 3.C** — Describe probability distributions.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.K** — Determine parameters of a sampling distribution for sample proportions. [Skill 3.B]
- **EK UNC-3.K.1** — For independent samples of a categorical variable from a population with proportion p , the sampling distribution of the sample proportion has mean $\mu = p$ and a standard deviation $\sigma = \sqrt{p(1-p)/n}$.
- **EK UNC-3.K.2** — If sampling without replacement, the standard deviation of the sample proportion is smaller than the formula above. If the sample size is less than 10% of the population size, the difference is negligible.
- **LO UNC-3.L** — Determine whether a sampling distribution for a sample proportion can be described as approximately normal. [Skill 3.C]
- **EK UNC-3.L.1** — For a categorical variable, the sampling distribution of the sample proportion will have an approximate normal distribution provided the sample size is large enough: $np \geq 10$ and $n(1-p) \geq 10$.
- **LO UNC-3.M** — Interpret probabilities and parameters for a sampling distribution for a sample proportion. [Skill 4.B]
- **EK UNC-3.M.1** — Probabilities and parameters for a sampling distribution for a sample proportion should be interpreted using appropriate units and within the context of a specific population.

Essential question: How do sampling, spread, and shape conditions limit what a surprising sample proportion can show?

DOK-3 skill: 4.B · **FRQ pattern:** failed-large-counts-normal-tail-vs-valid-binomial-and-bias

DOK rationale: Part (c) separates independence, shape, and unbiasedness to evaluate two overclaims and use a supplied probability to bound the conclusion supported by a random sample.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which condition supports treating the 100 selections as approximately independent?
- Why does $np = 8$ address shape rather than the estimator's long-run center?
- What values of X belong to $\hat{p} \geq 0.14$?
- How do 0.0135 and about 0.028 change the strength, but not the direction, of the evidence?

[WARN] LOOK FOR

- Applying an $n > 30$ rule carried over from sample means.
- Calling a nonnormal sampling distribution biased.
- Calculating exactly 14 rather than 14 or more caption users.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Uses the passing 10 percent condition and failed $np=8$ normal-shape condition to distinguish the two model requirements.
- **judgment-2** (required) — Uses supplied tail near 0.0282 as unusual evidence under $p=0.08$, without claiming proof of an increase.
- **judgment-3** (required) — Explains that SRS unbiasedness concerns center, whereas normality concerns shape, and rejects both overclaims.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

Common mistakes

- Checking only $n > 30$ instead of both expected counts
- Using $P(X = 14)$ rather than the upper-tail event $P(X \geq 14)$
- Saying the 10 percent condition determines normal shape rather than approximate independence and spread
- Equating a skewed sampling distribution with a biased estimator

[STAR] THE PROBLEM — annotated

Suppose a streaming platform has 5,000 active accounts and says that $p = 0.08$ of them regularly use captions. An SRS of 100 accounts, selected without replacement, contains 14 regular caption users, so $\hat{p} = 0.14$.

Jules: “Because $n = 100 > 30$, $\hat{p}$ is approximately normal. I get $P(\hat{p} \geq 0.14) = 0.0135$, which proves caption use increased.”

Nora: “Because $np = 8 < 10$, a normal model is not justified. Therefore $\hat{p}$ is biased and this sample tells us nothing.”

A binomial calculator gives $P(X \geq 14) = 0.0282$ for $X \sim \text{Bin}(100, 0.08)$. This closely approximates sampling from 5,000 accounts.

Caption-use sample

Source	Accounts	Users	Rate
Population	5,000	400	0.08
SRS	100	14	0.14

FIRST TAKE — one sentence before you work the parts

Does finding 14 caption users instead of the expected 8 prove that use increased? Give one reason.

Key: Not graded. Each student has one correct observation and one overreach.

[BOOK] CHECK INDEPENDENCE, THEN SHAPE (keep on the page)

For an SRS, $\mu_{\hat{p}} = p$ and $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$ when $n < 10\%$ of the population makes without-replacement dependence negligible. Approximate normality is a separate question: require both $np \geq 10$ and $n(1-p) \geq 10$; a generic $n > 30$ rule is not enough for proportions. If Large Counts fails, use an exact binomial calculation when independence is reasonable or simulate the sampling process.

DOK 1 (a) Identify p and verify the sample proportion $\hat{p} = 14/100$.

Key: The parameter is p , the proportion of all 5,000 active accounts that regularly use captions. The statistic is the sample proportion $\hat{p} = 14/100 = 0.14$. Thus $N = 5,000$ and $n = 100$.

DOK 2 (b) Find the mean and standard deviation of the sampling distribution of $\hat{p}$. Check the 10 percent condition and both Large Counts values.

Key: The sampling distribution has mean $\mu_{\hat{p}} = p = 0.08$ and standard deviation $\sigma_{\hat{p}} = \sqrt{0.08(0.92)/100} = 0.027129 \dots \approx 0.0271$. The 10 percent condition holds because $100 < 0.10(5,000) = 500$, so dependence from sampling without replacement is negligible. Large Counts does not fully hold: $np = 100(0.08) = 8 < 10$, while $n(1-p) = 100(0.92) = 92 \geq 10$.

DOK 3 (c) In 3–4 sentences, explain what each student gets right and wrong. Use the condition checks and the supplied probability 0.0282. State what the sample suggests about caption use and why a failed normal-shape check does not make the estimate biased.

Key: Neither claim is fully correct: the sample is random and $100 < 500$, but $np = 8 < 10$, so Nora correctly rejects the normal approximation. The supplied binomial tail, about 0.0282, makes 14 or more unusual under $p = 0.08$, giving evidence worth investigating rather than proof of an increase. The SRS produces an unbiased sample proportion; the failed condition concerns shape, not its long-run center. Jules overstates the evidence and Nora wrongly discards a useful estimate.